CIFAR-10 Image Classification using Convolutional Neural Network (CNN)

ABSTRACT:

This project presents the design, implementation, and evaluation of a Convolutional Neural Network (CNN) for image classification using the CIFAR-10 dataset. The objective is to automatically classify input images into one of ten object categories. The model learns hierarchical feature representations directly from raw pixel data, eliminating the need for manual feature engineering. Experimental results demonstrate that CNNs are highly effective for visual recognition tasks, achieving stable performance and good generalization.

INTRODUCTION:

Computer vision is a rapidly growing field of Artificial Intelligence that enables machines to interpret and understand visual data. Image classification is one of the fundamental tasks in this domain.

Traditional machine learning algorithms struggle with image data due to high dimensionality and spatial complexity. Deep Learning, particularly Convolutional Neural Networks (CNNs), addresses this challenge by automatically learning spatial hierarchies of features through convolutional filters.

This project implements a CNN model to classify CIFAR-10 images and evaluates its performance using standard metrics.

PROBLEM STATEMENT:

The goal of this project is to develop a supervised deep learning model capable of classifying input images into one of ten predefined categories using the CIFAR-10 dataset.

This is a multi-class classification problem where each image belongs to exactly one class.

DATASET DESCRIPTION:

The CIFAR-10 dataset is a benchmark dataset widely used in computer vision research.

Dataset Details:

- Total Images: 60,000 colored images
- Image Size: 32 × 32 pixels
- Channels: 3 (RGB)
- Number of Classes: 10

Classes:

Airplane, Automobile, Bird, Cat, Deer, Dog, Frog, Horse, Ship, Truck

Dataset Split:

- Training Set: 50,000 images
- Testing Set: 10,000 images

DATA PROCESSING:

Data preprocessing is a crucial step in improving model performance.

Steps performed:

- Pixel normalization: pixel values scaled from [0–255] to [0–1]
- Data reshaping ensured compatibility with CNN input layer
- Label encoding retained as integer values for sparse categorical loss
- Dataset integrity verified (no missing values or corrupted images)

EXPLORATORY DATA ANALYSIS(EDA):

Basic data exploration revealed:

- Balanced distribution across classes
- Images contain diverse object orientations and backgrounds
- Variations in lighting and color make classification challenging

- Some classes (e.g., cat vs dog) show visual similarity, increasing difficulty

CNN MODEL ARCHITECTURE:

A deep Convolutional Neural Network was designed to extract hierarchical features.

Architecture Details:

- Input Layer: 32 × 32 × 3 RGB images
- Convolution Layer 1:
  - Filters: 32
  - Kernel Size: 3×3
  - Activation: ReLU
- MaxPooling Layer 1:
  - Pool Size: 2×2
- Convolution Layer 2:
  - Filters: 64
  - Kernel Size: 3×3
  - Activation: ReLU
- MaxPooling Layer 2:
  - Pool Size: 2×2
- Convolution Layer 3:
  - Filters: 64
  - Kernel Size: 3×3
  - Activation: ReLU
- Flatten Layer
- Dense Layer:
  - Neurons: 64
  - Activation: ReLU
- Output Layer:
  - Neurons: 10
  - Activation: Softmax

MODEL COMPILATION:

The model was compiled using the following configuration:

- Optimizer: Adam (adaptive learning rate optimization)
- Loss Function: Sparse Categorical Crossentropy
- Metrics: Accuracy

Adam optimizer was chosen due to its efficiency in handling sparse gradients and faster convergence.

TRAINING PROCESS:

The model was trained for 10 epochs using the training dataset and validated using the test dataset.

Training Observations:

- Initial accuracy started around ~45%
- Gradual improvement observed across epochs
- Final training accuracy reached ~79%
- Validation accuracy stabilized around ~71–72%
- Loss decreased consistently, indicating proper learning

FUTURE ENHANCEMENT:

To enhance model performance, the following improvements can be implemented:

- Use of deeper architectures such as VGG16, ResNet
- Data augmentation techniques (rotation, flipping, zooming)
- Dropout regularization to reduce overfitting
- Hyperparameter tuning
- Transfer learning with pretrained models
- Deployment using Flask/FastAPI for real-world usage

TOOLS:

- Python
- TensorFlow / Keras

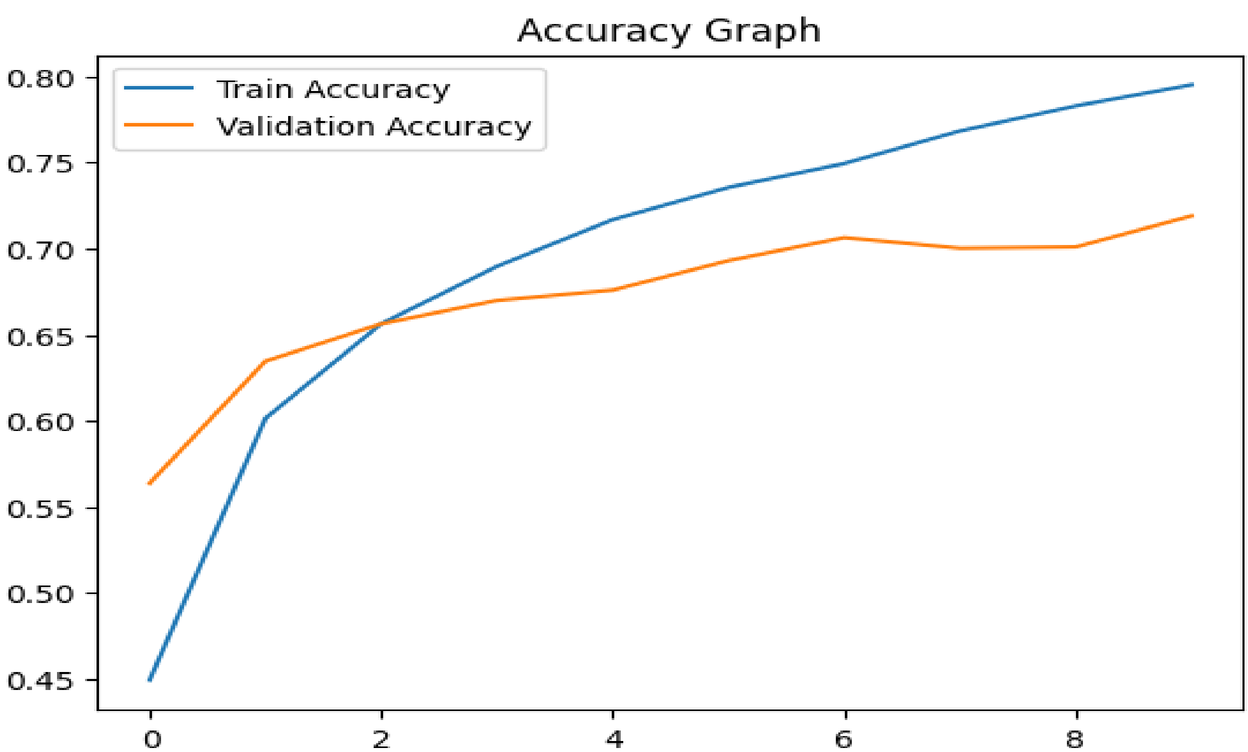
- NumPy
- Matplotlib
- Google Colab

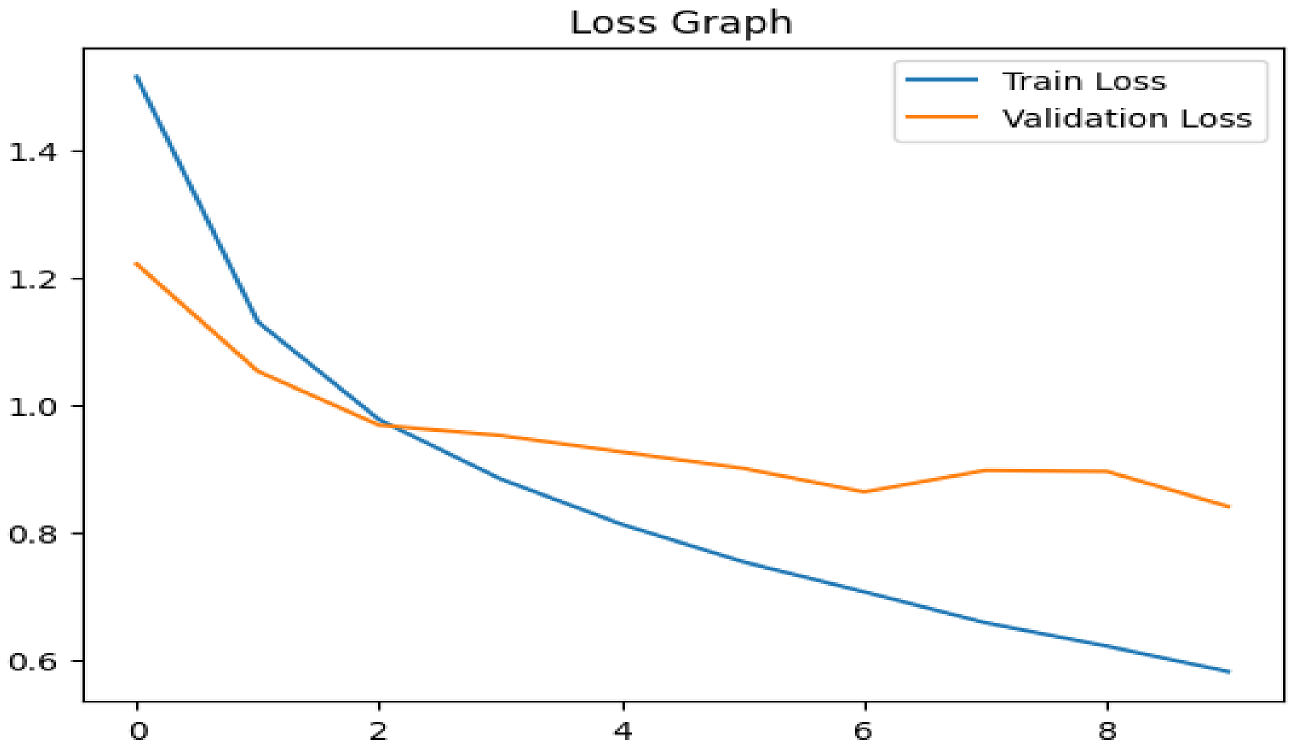
RESULT:

The Convolutional Neural Network (CNN) model was successfully trained on the CIFAR-10 dataset for image classification. The model performance was evaluated using training and validation accuracy along with loss metrics.

During training, the model showed consistent improvement in performance across epochs. The training accuracy increased gradually from around 45% to approximately 79%, indicating that the model effectively learned feature representations from the image data.

The validation accuracy stabilized around 71–72%, which shows that the model generalizes well to unseen data. The reduction in loss values over epochs further confirms that the model is learning effectively without major instability.





CONCLUSION:

This project successfully demonstrates the application of Convolutional Neural Networks for image classification using the CIFAR-10 dataset. The model achieved satisfactory performance with a validation accuracy of approximately 72%.

The results confirm that CNNs are powerful tools for visual recognition tasks and can be further improved with deeper architectures and advanced training techniques.